

A Drone Fleet-Borne SAR Model and Three-Dimensional Imaging Algorithm

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Abstract—Due to the advantages of low cost and flexible flight, small drones are showing tremendous application prospect in plenty of fields. Aiming at public area monitoring, a drone fleet-borne synthetic aperture radar (SAR) model is built in this paper. In this model, each radar is mounted on a small drone while illuminating the area of interest and the drones fly forward straightly in an array form. For practical feasibility, it is assumed that each radar works independently and simultaneously. To avoid interference, frequency-divided (FD) signals are adopted. Besides, bandwidth synthesizing is performed to improve the range resolution. Furthermore, 3D imaging is realized via combining the traditional 2D SAR imaging technique and the sensor array signal processing method. In specification, beamforming (BF) is adopted for the array direction compression. With flexible data acquisition model and fast imaging algorithm, 3D imaging using drone fleet borne SAR has great potential in real-time monitoring. The simulation experiments of point targets are performed and the imaging results validate the effectiveness of the proposed imaging model and 3D image formation algorithm.

Index Terms—Drone fleet, synthetic aperture radar (SAR), sensor array, frequency-divided (FD) signals, 3D imaging.

I. INTRODUCTION

DURING the past few decades, synthetic aperture radar (SAR) has shown great values in both military and civil fields [1]–[7], such as battlefield reconnaissance [1], [4], mapping [2], concealed weapon detection [3] and space debris tracking [5]. On the other hand, the rapid development of the small drone technology has caused much research attention, including monitoring and capturing towards black flight drones [8], [9], and studies on their potential applications [10]. At present, the main applications of small drones are mobile surveillance, aerial photography, etc.

Comparing with traditional large scale unmanned aerial vehicle (UAV), small drones have some remarkable advantages, such as flexible flight, low cost of hardware and easy manipulation. With these advantages, we can apply them in SAR, where the radar is mounted on the drone to actively

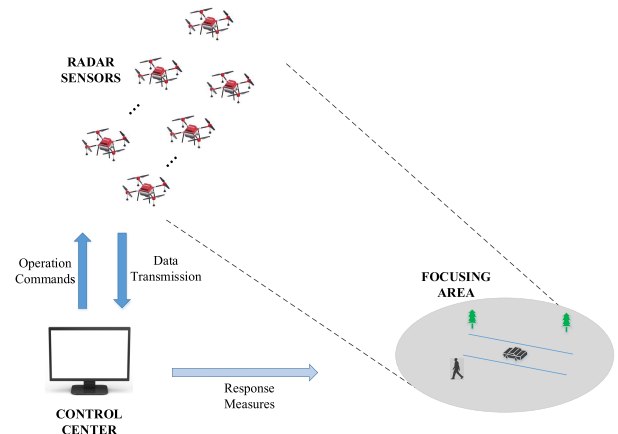


Fig. 1. Surveillance using drone fleet-borne radar sensors.

transmit electromagnetic wave towards the area of interest and simultaneously receive echoes reflected from the targets. Li *et al* carried out several imaging experiments with a drone-borne radar, where an ultra-wideband radar emitted pulses at a pulse repetition frequency (PRF) of 10MHz and a Wi-Fi connection was used to remotely control the radar [11], [12]. In these experiments, range profile analysis and two-dimensional (2D) imaging of an outdoor scene with trees and vehicles inside were implemented.

With small size and flexible flight, small drones can extend the application scope of SAR. Since the propagation of electromagnetic wave is generally not affected by the weather or air conditions, the drone-borne radar sensing is an effective supplement to the surveillance with cameras in security and protection systems. However, the load of a single drone is small, therefore the size and weight of the borne radar cannot be too large or too heavy, and this may limit the performances of the radar, such as the transmitting power, beam angle and bandwidth. To address these issues, the drone fleet can be adopted.

An example of surveillance using drone fleet-borne radar sensors can be shown in Fig. 1. In the surveillance, the control center gives commands to the flying path of the drones and signal transmitting orders of the sensors. At the same time, the radar receivers receive echoes from the scene and send them back to the signal processing center by wireless transmission. After postprocessing of the echoes, such as imaging, detection and recognition, response measurement can be given to the focusing area. Furthermore, if the received echo data is transmitted to the processing center in real-time with

Manuscript received May 14, 2019; revised June 11, 2019; accepted June 14, 2019. Date of publication June 17, 2019; date of current version September 18, 2019. This work was supported in part by the Project 2011 for Collaborative Innovation Center of Information Sensing and Understanding and in part by the Project for Innovation Special Zone of National Defense Science and Technology. The associate editor coordinating the review of this paper and approving it for publication was Laura Anitori. (Corresponding author: Jin Li.)

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Digital Object Identifier 10.1109/JSEN.2019.2923529

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wireless communication, such as the upcoming 5G, the real-time surveillance and fast reaction is possible, and it would be a promising application in the internet of things (IoT).

In SAR imaging, 3D imaging has become a research focus for its capability of providing more information and details of the targets [5], [13]–[15]. As far, plenty of 3D SAR imaging models and image formation algorithms have been proposed [16]–[25]. Among these studies, both the linear array [18]–[23] model and the planar array [24], [25] model are popular for their ability to obtain high resolution in all the three dimensions. In this paper, a drone fleet-borne SAR model with sensor array is built and 3D imaging is realized via combining the traditional 2D imaging and the sensor array signal processing.

Generally, for the array SAR, all the sensors are fixed on one vehicle, thus the same signal form are adopted and the signals of each sensor can be distinguished through a transmitting time sequence [23], [26]. However, for the drone fleet-borne sensors, it is difficult to precisely control this sequence. In reality, it is more practical to use the all-transmitting-all-receiving pattern, where each drone-borne sensor transmits signal and receives echo independently. In this case, to solve the problem of signal interference, the frequency-divided (FD) signals are used in this paper [27]–[29]. Moreover, with FD signals, larger bandwidth can be obtained through bandwidth synthesizing and larger bandwidth leads to higher range resolution.

With broadband signal and synthetic aperture, each sensor can form the 2D image using the traditional SAR imaging algorithms, such as the Range-Doppler algorithm (RDA) and back-projection algorithm (BPA). Since the computation speed of RDA is much faster than that of BPA, RDA is more suitable for real-time imaging. In cross-track direction, the sensor array form the aperture for imaging and the beamforming (BF) technique is employed to realize the array direction compression [30].

The remaining part of this paper is organized as follows. Section II introduces the imaging geometry and the signal model. In section III, 3D imaging algorithm is presented, which consists of the signal separation, bandwidth synthesizing and 3D image formation procedure. Section IV gives the numerical simulation experiment to verify the effectiveness of the proposed SAR model and 3D imaging algorithm. Conclusion is presented in section V.

II. IMAGING GEOMETRY AND SIGNAL MODEL

A. Imaging Geometry

For ground area surveillance, such as road traffic detection and building scanning, the downward-looking model is suitable for SAR imaging. The imaging geometry is shown in Fig. 2, where the three orthogonal dimensions of X, Y and Z indicate the along-track/azimuth, cross-track and height/range directions, respectively. In this model, the drone fleet align in a row, which parallels with Y-axis, and fly forward straightly along X-axis with a constant velocity of v . The height of the drone fleet is set as H . We assume that the drones distribute uniformly with an internal spacing d during the flight.

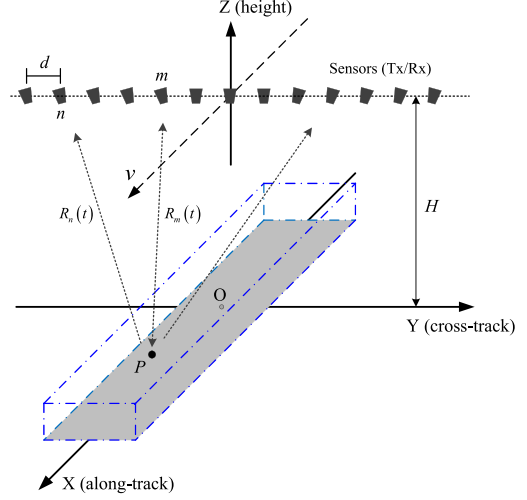


Fig. 2. Imaging geometry of the drone fleet-borne sensors. The three dimensions X, Y and Z correspond to the along-track/azimuth, cross-track and height/range directions, respectively.

In this model, the sensor array form an aperture in the cross-track direction and the motion of the drones form a synthetic aperture in the along-track direction. At slow time t , the instantaneous location of the m th sensor is (vt, l_m, H) , $m = 1, 2, \dots, M$, where M is the number of the radar sensors, $l_m = (m - \frac{M+1}{2})d$ is the coordinate in Y-axis.

To be more feasible in practice, the fully transmitting-receiving scheme is adopted, which indicates that all of the sensors transmit electromagnetic waves and receive echoes independently. Thus, the number of receivers N is equal to that of transmitters M ($M = N$). In this scheme, each drone-borne radar has its transceiver module, which allows it to work independently or cooperate with other sensors to form a fleet, thus making the usage of the drones more flexible. Besides, if some of the drones break down in the fleet, the others can realign to work.

B. FD-LFM Signal Model

Linear frequency modulated (LFM) signal has been widely used in SAR system for its broad band and implementation simplicity. To avoid signal interference and for the convenience of separation in the following processing, the FD-LFM signals are adopted. For the m th sensor, the transmitted signal is expressed as

$$s_m(\tau) = \text{rect}\left(\frac{\tau}{T_s}\right) \exp\left[j2\pi\left(f_m\tau + \frac{1}{2}\gamma_m\tau^2\right)\right] \quad (1)$$

where τ is the fast time, T_s is the pulse duration and supposed to be the same for all the transmitted signals, f_m and γ_m are the carrier frequency and chirp rate, respectively. For simplicity, we suppose that each signal has the same chirp rate, therefore, γ is denoted as the chirp rate in the following for brevity. $\text{rect}(\cdot)$ is a rectangular window function defined as

$$\text{rect}\left(\frac{\tau}{T_s}\right) = \begin{cases} 1, & \tau \leq \frac{T_s}{2} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

According to the analysis in [28], the correlation of two FD-LFM signals depends much on the difference of their

frequency centers. It demonstrates that the mutual correlation of two different FD-LFM signals is much lower than one's autocorrelation. Therefore, separation could be achieved for the following processing of the radar echoes.

Besides, in [31], it proves that the selection of frequency step can influence the sidelobe and suppression effect. When the frequency step is smaller than the single bandwidth, the sidelobe can be weakened, but the suppression effect is not ideal, and vice versa. As a compromise, the frequency step equaling to the single bandwidth is applied, which indicates that $f_m - f_{m-1} = B_s$, where B_s is the single bandwidth and satisfies $B_s = \gamma T_s$. Hence, the center frequency of the m th signal is given by

$$f_m = f_0 + (m - 1) B_s \quad m = 1, 2, \dots, M \quad (3)$$

where f_0 is the center frequency of the first signal.

C. Echo Model

Supposing that an arbitrary point scatterer P with coordinate (x_P, y_P, z_P) locates in the imaging scene, as shown in Fig. 2, the instantaneous distance from the m th sensor to point scatterer P at slow time t is given by

$$R_m(t; P) = \sqrt{(x_P - vt)^2 + (y_P - l_m)^2 + (z_P - H)^2} \quad (4)$$

Generally, the signal transmitted by the transmitter m travels to the target P and returns to the receiver n ($n = 1, 2, \dots, N$), thus the round trip distance is expressed as

$$R_{mn}(t; P) = R_m(t; P) + R_n(t; P) \quad (5)$$

In addition, the point scatterer P receives signals from all of the transmitters and reflects them to each receiver. Therefore, the received echo of the n th ($n = 1, 2, \dots, N$) receiver from P is expressed as

$$s_n(\tau, t; P) = \sum_{m=1}^M \sigma_P s_m \left[\tau - \frac{R_{mn}(t; P)}{c} \right] \quad (6)$$

where σ_P is the reflectivity of P and c is the speed of light.

Besides, based on the first-order Born approximation theory, the whole scene can be expressed as the linear superposition of all the point scatterers. Therefore, the received echo is given as

$$s_n(\tau, t) = \sum_{q=1}^Q \sum_{m=1}^M \sigma_q s_m \left[\tau - \frac{R_{mn}(t; q)}{c} \right] \quad (7)$$

where Q is the number of scatterers in the imaging scene.

III. 3D IMAGING ALGORITHM

A. Signal Separation

With N receivers, N echo data matrices are obtained and each one contains the signals from all the transmitters. For the following processing, we need to separate each transmitter's echo from the N echo matrices. To implement separation, multiple frequency mixers and the low-pass (LP) filter are applied. The separation process is depicted in Fig. 3.

After separation, the sub-echo corresponding to the m th-transmitter- n th-receiver is obtained. Without loss of generality

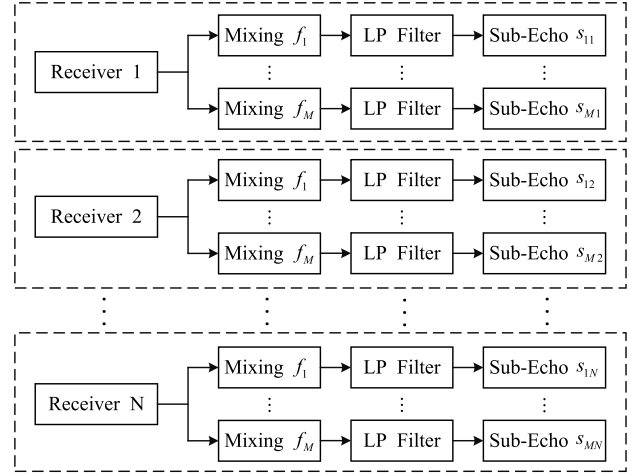


Fig. 3. Signal separation process.

and for writing convenience, we denote (6) as the echo model. Then, the separated sub-echo is given by

$$s_{mn}(\tau, t; P) = \sigma_P \text{rect} \left(\frac{\tau - R_{mn}(t; P)/c}{T_s} \right) \times \exp \left\{ -j \frac{2\pi f_m}{c} R_{mn}(t; P) \right\} \times \exp \left\{ j\pi \gamma \left[\tau - \frac{R_{mn}(t; P)}{c} \right]^2 \right\} \quad (8)$$

From Fig. 3, we know that MN sub-echoes are obtained after the separation. According to the phase center approximation (PCA) theory, the bistatic sensors can be equivalent as a virtual monostatic one, which is located at the center of the two [32], [33]. Therefore, a virtual sensor located at $(vt, \frac{l_m + l_n}{2}, H)$ at slow time t is assumed for the m th-transmitter- n th-receiver pair.

However, this approximation may lead to phase errors. According to the analysis in [34], the errors depend on several factors, including the interval between the real sensors, the distance from the target to the virtual sensors and the signal wavelength. In [32], A. Belletini *et al.* declared that the errors can be neglected if the double wave-path difference is less than a quarter-wavelength, which indicates the far field condition. Otherwise, they need to be compensated before the following processing and a compensation method is shown in [35].

The distance between the equivalent virtual sensor of m th-transmitter- n th-receiver and P is denoted as $R_{e,mn}(t; P)$. Once the phase errors are compensated, the round trip distance $R_{mn}(t; P)$ in (8) can be replaced by $2R_{e,mn}(t; P)$.

With M real sensors displaced uniformly, $2M - 1$ virtual sensors can be obtained. A Demonstration of the relationship between the real and virtual sensors is shown in Fig. 4, where there are 4 real sensors corresponding to 7 virtual ones.

Since $l_m = (m - \frac{M+1}{2})d$, $l_n = (n - \frac{N+1}{2})d$ and $M = N$, the Y-axis coordinate of the virtual sensor corresponding to the m th-transmitter- n th-receiver is $\frac{l_m + l_n}{2} = \frac{m+n-(M+1)}{2}d$. Therefore, there exist some (m, n) pairs that are corresponded to the same virtual sensor. This can also be seen in Fig. 4. Take the virtual sensor iv as an example, it corresponds to

the real sensor pairs (I, IV), (II, III), (III, II) and (IV, I). Thereupon, the echoes corresponding to the same virtual sensor are redundant for the following processing. Additionally, the redundancy makes the imaging process not be impacted greatly even if some of the drones break down.

B. Bandwidth Synthesizing and Range Compression

Since redundant echoes are obtained for the same virtual sensor, we can make full use of them to improve the imaging quality. Thus, bandwidth synthesizing is carried out to expand the signal bandwidth, which can further increase the range resolution.

Firstly, frequency shifting is operated to (8) with a shifting function

$$u_{1,m}(\tau) = \exp\{j2\pi B_s(m-1)\tau\} \quad (9)$$

Besides, after the virtual sensor equivalence, we can replace $R_{mn}(t; P)$ with $2R_{e,mn}(t; P)$ in (8). Then, by multiplying (8) with (9), we can get

$$\begin{aligned} s_{1,mn}(\tau, t; P) &= \sigma_{prect}\left(\frac{\tau - 2R_{e,mn}(t; P)/c}{T_s}\right) \\ &\times \exp\left\{-j\frac{4\pi f_0}{c}R_{e,mn}(t; P)\right\} \\ &\times \exp\left\{j\pi\gamma\left[\tau - \frac{2R_{e,mn}(t; P)}{c} + (m-1)T_s\right]^2\right\} \\ &\times \exp\left\{-j\pi\gamma[(m-1)T_s]^2\right\} \end{aligned} \quad (10)$$

The last exponential term in (10) is removed with a function

$$u_{2,m} = \exp\left\{j\pi\gamma[(m-1)T_s]^2\right\} \quad (11)$$

To implement bandwidth synthesizing, we transform the signal to frequency domain. Thus, Fourier Transform (FT) is performed with respect to fast time τ and the derivation is presented in Appendix V. Then, the range frequency domain signal is given as

$$\begin{aligned} S_{1,mn}(f_r, t; P) &= A_1 \sigma_{prect}\left[\frac{f_r - (m-1)B_s}{B_s}\right] \times \exp\left\{-j\frac{\pi}{\gamma}f_r^2\right\} \\ &\times \exp\left\{-j\frac{4\pi}{\lambda}R_{e,mn}(t; P)(f_r + f_0)\right\} \\ &\times \exp\{j2\pi(m-1)T_s f_r\} \end{aligned} \quad (12)$$

where A_1 is the amplitude after FT, which does not effect the signal phase.

The last exponential term in (12) is removed with a function

$$u_{3,m}(f_r) = \exp\{-j2\pi(m-1)T_s f_r\} \quad (13)$$

and we can get

$$\begin{aligned} S_{2,mn}(f_r, t; P) &= A_1 \sigma_{prect}\left[\frac{f_r - (m-1)B_s}{B_s}\right] \times \exp\left\{-j\frac{\pi}{\gamma}f_r^2\right\} \\ &\times \exp\left\{-j\frac{4\pi}{c}R_{e,mn}(t; P)(f_r + f_0)\right\} \end{aligned} \quad (14)$$

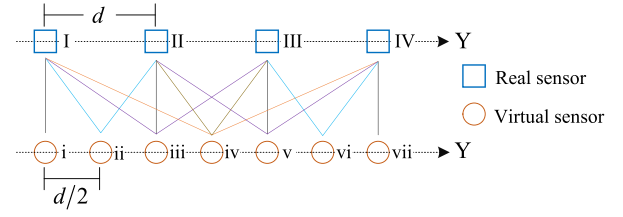


Fig. 4. Demonstration of the real and virtual sensors.

Supposing that there are M_1 pairs of echoes that are corresponding to the same virtual sensor, the bandwidth synthesizing signal is given by

$$\begin{aligned} S_3(f_r, t; P) &= \sum_{m=m_1}^{m_1+M_1-1} S_{2,(m)}(f_r, t; P) \\ &= A_1 \sigma_{prect}\left[\frac{f_r - M_1 B_s/2}{M_1 B_s}\right] \exp\left\{-j\frac{\pi}{\gamma}f_r^2\right\} \\ &\times \exp\left\{-j\frac{4\pi}{c}R_{e,(m)}(t; P)(f_r + f_0)\right\} \end{aligned} \quad (15)$$

where (m) indicates the (m, n) pair corresponding to the same virtual sensor and m_1 is the starting number.

It should be noted that M_1 is different for different virtual sensors, which can also be seen in Fig. 4. However, from (15), we know that this difference only influences the synthesized bandwidth and would not affect the following processing.

Matched filtering (MF) is performed to (15) to achieve range compression and inverse FT (IFT) is taken with respect to frequency f_r , then we can obtain

$$\begin{aligned} s_{(m)}(\tau, t; P) &= A_2 \sigma_P \sin c\left\{M_1 B_s\left[\tau - \frac{2R_{e,(m)}(t; P)}{c}\right]\right\} \\ &\times \exp\left\{-j\frac{4\pi f_0}{c}R_{e,(m)}(t; P)\right\} \end{aligned} \quad (16)$$

where A_2 is the amplitude after IFT, $\sin c(\cdot)$ denotes the sinc function defined as $\sin c(t) = \frac{\sin \pi t}{\pi t}$.

So far, the bandwidth synthesizing and range compression have been completed, and the bandwidth of the synthesizing signal has increased from B_s to $M_1 B_s$. Besides, from (16), we can see that the center frequency for each virtual sensor has been aligned to f_0 .

C. Along-Track Compression

After range compression, the range cell migration (RCM) emerges due to the motion of the sensors and it could impact the compression in along-track direction and defocus the image. Therefore, the range cell migration correction (RCMC) should be completed before along-track compression. Generally, because the original range cells and the desired range cells are known, RCMC is realized through interpolation [36].

From Fig. 4, we know that the internal distance of the virtual sensors is $d/2$. Since there are $2M-1$ virtual sensors, we set the sequence number of the virtual sensor as β , $\beta = 1, \dots, 0, \dots, 2M-1$, then the coordinate of the virtual sensor is (vt, y_β, H) , where $y_\beta = \frac{\beta-M}{2}d$. Thus, the distance between the virtual sensor β and P is given by

$$R_\beta(t; P) = \sqrt{(x_P - vt)^2 + (y_P - y_\beta)^2 + (z_P - H)^2} \quad (17)$$

Taylor expansion of (17) is taken with respect to slow time t and the cubical and higher order terms can be neglected with the condition that $H \gg x_P, y_P, z_P$, which is satisfied in general. Then, the distance is presented as

$$R_\beta(t; P) \approx R_P + \frac{(x_P - vt)^2}{2R_P} + \frac{y_\beta^2 - 2y_P y_\beta}{2R_P} \quad (18)$$

where $R_P = \sqrt{y_P^2 + (z_P - H)^2}$ is the projection of $R_\beta(t; P)$ on plane YOZ.

Substituting $R_{e,(m)}(t; P)$ with $R_\beta(t; P)$ in (16), we can get the range compressed signal of the virtual sensor β as

$$\begin{aligned} s_\beta(\tau, t; P) = & A_2 \sigma_P \sin c \left\{ M_1 B_s \left[\tau - \frac{2R_\beta(t; P)}{c} \right] \right\} \\ & \times \exp \left\{ -j \frac{2\pi}{\lambda R_P} [y_\beta^2 - 2y_P y_\beta] \right\} \\ & \times \exp \left\{ -j \frac{2\pi}{\lambda R_P} (x_P - vt)^2 \right\} \\ & \times \exp \left\{ -j \frac{4\pi}{\lambda} R_P \right\} \end{aligned} \quad (19)$$

where $\lambda = \frac{c}{f_0}$ is the wavelength.

In (19), the first exponential term is relevant to the cross-track direction, while the second exponential term contains the along-track information. The third exponential term is a constant. Besides, y_β is known for a fixed virtual sensor and thus can be compensated.

The along-track compression can also be realized by MF and the matched filter is given by

$$\Phi(t) = \exp(-j\pi f_d t^2) \quad (20)$$

where $f_d = -\frac{2v^2}{\lambda H}$ is Doppler frequency modulated rate in along-track direction.

After the along-track compression, we can get the signal as

$$\begin{aligned} s_{4,\beta}(\tau, t; P) = & A_3 \sigma_P \sin c \left\{ M_1 B_s \left[\tau - \frac{2R_\beta(t; P)}{c} \right] \right\} \\ & \times \sin c \left\{ B_a \left(t - \frac{x_P}{v} \right) \right\} \cdot \exp \left\{ -j \frac{4\pi}{\lambda} R_P \right\} \\ & \times \exp \left\{ j \frac{4\pi y_\beta y_P}{\lambda R_P} \right\} \end{aligned} \quad (21)$$

where A_3 is the amplitude after along-track compression, B_a is the Doppler bandwidth in along-track direction.

D. Cross-Track Compression

In cross-track direction, the virtual sensor array forms the aperture for imaging. In (21), the last exponential term contains the cross-track phase. To realize the compression in cross-track direction, BF technique is adopted for its advantages of improving the signal-to-noise rate (SNR) and suppressing the ambiguity [37], [38].

In (21), let $\sin \eta = \frac{y_P}{R_P}$, where η is the pitch angle of P in the YOZ plane. BF method is to weight the signal in (21)

with different weighting factors. The weighting result is given by

$$\begin{aligned} s_{5,\beta}(\tau, t, k; P) = & s_{4,\beta}(\tau, t; P) \exp \left\{ -j \frac{4\pi y_\beta}{\lambda} \sin \varphi_k \right\} \\ = & A_3 \sigma_P \sin c \left\{ M_1 B_s \left[\tau - \frac{2R(t, \beta; P)}{c} \right] \right\} \\ & \times \sin c \left\{ B_a \left(t - \frac{x_P}{v} \right) \right\} \times \exp \left\{ -j \frac{4\pi}{\lambda} R_P \right\} \\ & \times \exp \left\{ j \frac{4\pi y_\beta}{\lambda} (\sin \eta - \sin \varphi_k) \right\} \end{aligned} \quad (22)$$

where φ_k is the divided angle in the cross-track direction and $k = 1, 2, \dots, K$, where K is the total divided number.

After the weighing, the signal data corresponding to the β th sensor has been expanded from a 2D matrix to a 3D matrix. Furthermore, we add all the $2M - 1$ 3D data matrix and the result is given by

$$\begin{aligned} s_6(\tau, t, k; P) = & \sum_{\beta=1}^{2M-1} s_{5,\beta}(\tau, t, k; P) \\ = & A_3 \sigma_P \sin c \left\{ M_1 B_s \left[\tau - \frac{2R(t, \beta; P)}{c} \right] \right\} \\ & \times \sin c \left\{ B_a \left(t - \frac{x_P}{v} \right) \right\} \times \exp \left\{ -j \frac{4\pi}{\lambda} R_P \right\} \\ & \times \frac{\sin \left\{ \frac{(2M-1)\pi d}{\lambda} (\sin \eta - \sin \varphi_k) \right\}}{\sin \left\{ \frac{\pi d}{\lambda} (\sin \eta - \sin \varphi_k) \right\}} \end{aligned} \quad (23)$$

Since $\frac{\pi d}{\lambda} (\sin \eta - \sin \varphi_k)$ is generally small, we have $\sin \left\{ \frac{\pi d}{\lambda} (\sin \eta - \sin \varphi_k) \right\} \approx \frac{\pi d}{\lambda} (\sin \eta - \sin \varphi_k)$. Then, (23) can be rewritten as

$$\begin{aligned} s_6(\tau, t, k; P) \approx & A_3 \sigma_P \sin c \left\{ M_1 B_s \left[\tau - \frac{2R(t, \beta; P)}{c} \right] \right\} \\ & \times \sin c \left\{ B_a \left(t - \frac{x_P}{v} \right) \right\} \times \exp \left\{ -j \frac{4\pi}{\lambda} R_P \right\} \\ & \times (2M - 1) \sin c \left\{ \frac{(2M - 1)d}{\lambda} (\sin \eta - \sin \varphi_k) \right\} \end{aligned} \quad (24)$$

Now that the cross-track compression has been completed, we get the final 3D image of point scatterer P . Furthermore, the imaging results of the whole scene can be presented as

$$s_7(\tau, t, k) = \sum_{q=1}^Q s_6(\tau, t, k; q) \quad (25)$$

The 3D imaging procedure is presented in Fig. 5.

E. Resolution Analysis

1) *Range Resolution*: The range resolution is decided by the bandwidth of the synthesized signal and is given by

$$\rho_r = \frac{c}{2M_1 B_s}. \quad (26)$$

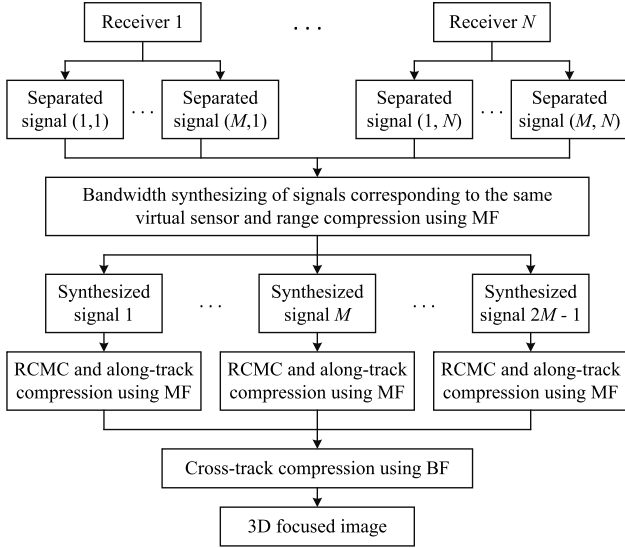


Fig. 5. flowchart of the 3D imaging algorithm.

It can be seen that the resolution has increased by M_1 comparing with that using a single echo.

2) *Along-Track Resolution*: Like the traditional 2D SAR imaging, the along-track resolution is decided by the true aperture of the radar antenna D_a , which is given by

$$\rho_a = \frac{D_a}{2} \quad (27)$$

3) *Cross-Track Resolution*: The angle resolution in cross-track direction is determined by the wavelength and the length of the cross-track aperture, which is given by

$$\rho_\eta = \frac{\lambda}{(M-1)d} \quad (28)$$

Therefore, the space resolution is presented as $\rho_c = \rho_\eta R_0$, where R_0 is the distance from the central line to the sensor array.

IV. SIMULATION RESULTS AND ANALYSIS

To verify the effectiveness of the proposed imaging model and 3D image formation algorithm, numerical simulations are carried out in this section. The radars have different carrier frequencies and the first center frequency is set as 88GHz. Bandwidth of each radar is assumed to be equal, which is 300MHz in this simulation. The spacing of the adjacent drones is set as 0.5m for realizability. Since this spacing is much larger than the half-wavelength, we need to control the scanning angle of the radar sensors to avoid grating lobes. Therefore, for the application of road surveillance, the cross-track width of the imaging area is assumed to be 20m and the height of the drones is set as 150m. Besides, the beam angle of a single radar in along-track direction is 1° and the synthetic aperture at the ground plane can be computed, which approximates to 2.6m. To avoid Doppler ambiguities in along-track direction, the PRF of the radars should be greater than the Doppler bandwidth, thus it is set as 300Hz in this simulation. The main parameters of the radar system and the drone fleet layout are listed in Table I.

TABLE I
MAIN PARAMETERS

Parameters	Values
Bandwidth B_s (MHz)	300
Time duration T_s (μ s)	1
PRF (Hz)	300
Number of drones M	41
Internal spacing d (m)	0.5
Height of drones H (m)	150
Velocity of drones v (m/s)	15

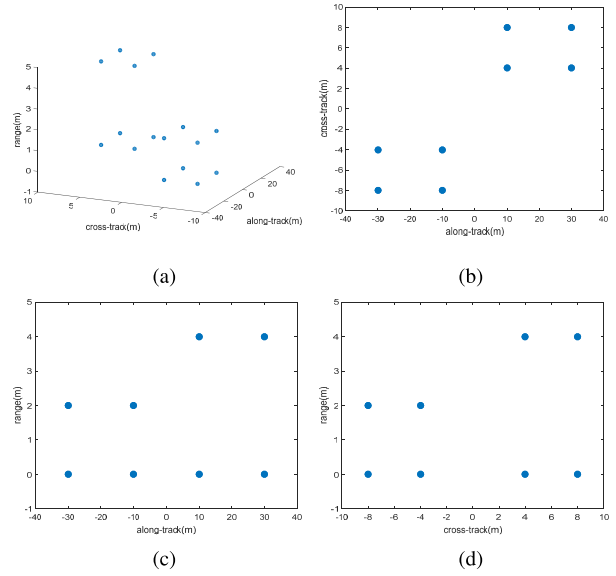


Fig. 6. Simulated point scatterers. (a) 3D scene. (b) XOY plane (along-track-cross-track) view. (c) XOZ plane (along-track-height) view. (d) YOZ plane (cross-track-height) view.

The target is simulated with 16 point scatterers, which have different coordinates in the 3D space, as shown in Fig. 6. Fig. 6(a) is the 3D scene of the point scatterers and Fig. 6(b)-(d) are the three views of the scene, respectively. The imaging results using the proposed 3D image formation method are shown in Fig. 7. Fig. 7(a) is the 3D scene of the reconstructed point scatterers and Fig. 7(b)-(d) are the three views of the 3D image, respectively. Comparing Fig. 6 with Fig. 7, it can be seen that the point scatterers are well reconstructed.

Furthermore, to verify the effectiveness of the bandwidth synthesizing, the range/height profiles at the along-track-cross-track cell $(-30\text{m}, -8\text{m})$ are plotted in Fig. 8, where the red solid line and the blue dashed line correspond to the compressed results with and without bandwidth synthesizing, respectively. In Fig. 8, five signals that correspond to the same virtual sensor are synthesized. Fig. 8(b) is the zoomed in image of the green region in Fig. 8(a) and the black solid line corresponds to the level of 0.707, where the width at this level is generally seen as the resolution. It can be seen that the main lobe of the signal with bandwidth synthesizing is nearly five times narrower than that of the one without bandwidth synthesizing. It should be noted that the number of bandwidth synthesizing signals may be different for each virtual sensor in this paper, therefore the reduction degree of the main lobe

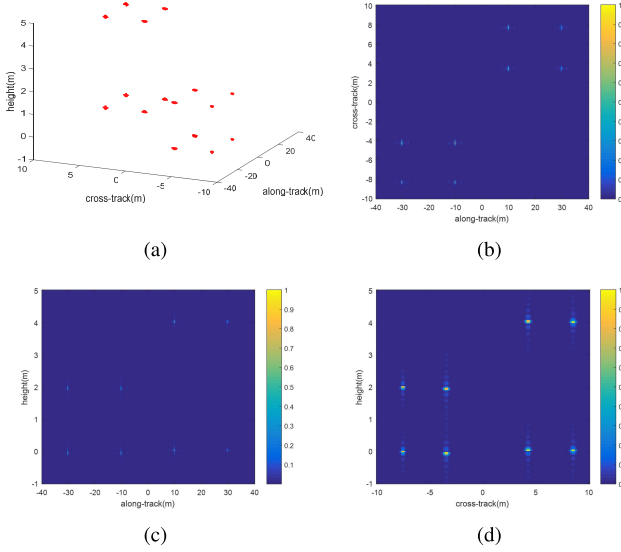


Fig. 7. 3D imaging results. (a) Imaging 3D scene. (b) XOY plane (along-track-cross-track) view. (c) XOZ plane (along-track-height) view. (d) YOZ plane (cross-track-height) view.

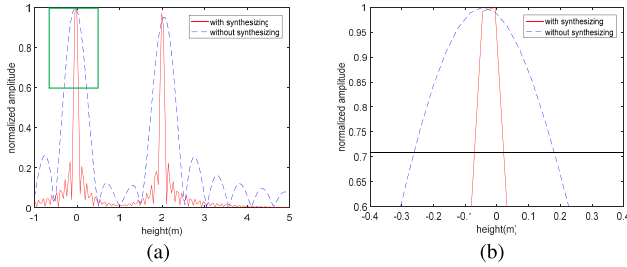


Fig. 8. (a) Height profiles at along-track-cross-track cell (-30m,-8m). (b) Zoomed in image of the green region in (a).

after the range compression varies, but it would not affect the resolution in along-track and cross-track directions.

V. CONCLUSION

Aiming at the researches of potential applications of the small drones, a SAR imaging scheme using drone fleet-borne radar sensors is proposed in this paper. In this scheme, each radar sensor is mounted on a flying drone and all the drones align in a row. To avoid signal interference, FD signals are adopted and the signal separation procedure is introduced. Furthermore, bandwidth synthesizing is performed to make full use of the echoes and so as to improve the range resolution. By combining the traditional 2D SAR imaging technique and the sensor array signal processing method, 3D imaging of the area of interest is achieved. Numerical simulation experiments are performed and the 3D imaging ability of the proposed model and the image formation algorithm are verified.

However, the perturbation of the drone fleet's motion is not considered in this paper, and it would affect the imaging quality in the practical application. Besides, synchronization is another important issue to be considered. Thus, they need to be studied in the future research. In addition, carrying out the practical experiments is also a focus in the following work.

APPENDIX

Let $\tau_0 = \frac{2R_{e,mn}(t;P)}{c}$, the terms relating to variable τ in (10) can be extracted as

$$s_1(\tau) = \text{rect}\left(\frac{\tau - \tau_0}{T_s}\right) \exp\left\{j\pi\gamma[\tau - \tau_0 + (m-1)T_s]^2\right\} \quad (\text{A1})$$

By expanding (A1), we can get

$$\begin{aligned} s_1(\tau) &= \text{rect}\left(\frac{\tau - \tau_0}{T_s}\right) \exp\left\{j\pi\gamma(\tau - \tau_0)^2\right\} \\ &\quad \times \exp\{j2\pi B_s(m-1)(\tau - \tau_0)\} \\ &\quad \times \exp\left\{j\pi\gamma[(m-1)T_s]^2\right\} \end{aligned} \quad (\text{A2})$$

Rewrite the first two terms in (A2) as $s_{11}(\tau) = \text{rect}\left(\frac{\tau - \tau_0}{T_s}\right) \exp\{j\pi\gamma(\tau - \tau_0)^2\}$ and the first three terms as $s_{12}(\tau) = s_{11}(\tau) \cdot \exp\{j2\pi B_s(m-1)(\tau - \tau_0)\}$. It is known that the FT of the LFM signal $s(\tau) = \text{rect}\left(\frac{\tau}{T_s}\right) \exp(j\pi\gamma\tau^2)$ is given by

$$S(f) = \text{rect}\left(\frac{f}{B_s}\right) \exp\left\{-j\frac{\pi}{\gamma}f^2\right\} \quad (\text{A3})$$

Combining the property that

$$FT_{(\tau)}[s(\tau - \tau_0)] = \exp(-j2\pi f\tau_0) S(f) \quad (\text{A4})$$

we can get the FT of $s_{11}(\tau)$ as

$$S_{11}(f) = \text{rect}\left(\frac{f}{B_s}\right) \exp\left\{-j\frac{\pi}{\gamma}f^2\right\} \exp\{-j2\pi f\tau_0\} \quad (\text{A5})$$

Furthermore, in combination with the property that

$$FT_{(\tau)}\left[e^{j2\pi f_0\tau}s(\tau)\right] = S(f - f_0) \quad (\text{A6})$$

we can get the FT of $s_{12}(\tau)$ as

$$\begin{aligned} S_{12}(f) &= \text{rect}\left[\frac{f - (m-1)B_s}{B_s}\right] \exp\{-j2\pi f\tau_0\} \\ &\quad \times \exp\left\{-j\pi\frac{[f - (m-1)B_s]^2}{\gamma}\right\} \end{aligned} \quad (\text{A7})$$

By multiplying the last term in (A2), the FT of $s_1(\tau)$ is obtained as

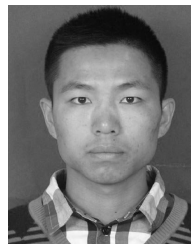
$$\begin{aligned} S_1(f) &= \text{rect}\left[\frac{f - (m-1)B_s}{B_s}\right] \exp\{-j2\pi f\tau_0\} \\ &\quad \times \exp\left\{-j\frac{\pi}{\gamma}f^2\right\} \exp\{j2\pi(m-1)T_sf\} \end{aligned} \quad (\text{A8})$$

At last, multiplying $S_1(f)$ with other terms in (10), we can obtain (12).

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